Filter-Induced and Learned Deadlocks in Safety-Filtered Reinforcement Learning for Mobile Robot Navigation

Pouya Bathaei Pourmand

Draft, October 26, 2026

Abstract

A control-barrier-function (CBF) safety filter can make a learned navigation policy collision-free, but the robot may then stall in front of an obstacle and time out. We analyse such deadlocks for a unicycle robot whose speed is limited by a one-step discrete-time CBF filter. We first characterise the filter geometrically. Its one-step safety guarantee does not depend on the commanded turn rate, so any modification of the turn rate preserves the guarantee. Near the keep-out boundary, the set of headings that permit motion shrinks to the half-plane that points away from the obstacle. We then classify the timeouts of ten proximal-policy-optimisation (PPO) agents and two classical controllers on 300 seeded test scenarios. In every scenario, a geometric oracle confirms that a path keeping the safety margin exists. Two mechanisms dominate, and they depend on how the agent was trained. Agents trained without the filter and run with it are *blocked by the filter* at the boundary (141 of 175 expert timeouts). Agents trained with the filter mostly *stop by themselves* before the boundary (122 of 195), so the filter is not the immediate cause. In paired comparisons on the expert stage, training with the filter lowered the filter-blocked rate by 14.8 percentage points (95 % interval [5.6, 24.4]) and raised the self-stop rate by 17.6 points [8.2, 27.0]. Sampling actions instead of using the policy mean removed most self-stops but no filter-blocked stalls. Both training contrasts replicated in a pre-registered evaluation on 500 previously unused scenarios. In the same evaluation, a test-time heading modification that cannot weaken the safety guarantee lowered the filter-blocked rate of agents trained without the filter by 7.4 points [4.8, 10.2] and raised their success by 7.2 points [5.0, 9.7], with zero collisions. It left self-stops unchanged. Self-stops did not disappear with longer training: they were more frequent at the final training checkpoints than at the selected ones. All results are from simulation with exact state and static circular obstacles, and they do not establish real-world safety. **[EVIDENCE REQUIRED:** Results of the correction-penalty baseline (E1) must be added before submission; see minimum_experiments.md.]

Keywords: safe reinforcement learning; control barrier functions; safety filter; deadlock; mobile robot navigation; liveness.

1 Introduction

Safety filters are a common way to make learned controllers usable in safety-critical settings [1, 2]. The learned policy proposes an action, and a filter, often a control barrier function (CBF) quadratic program [3, 4], replaces it with the closest action that keeps the system in a safe set. In robot navigation, the safe set excludes a margin around every obstacle, and the filter guarantees under its modelling assumptions that the robot never enters that margin.

The guarantee says nothing about reaching the goal. For model-based nominal controllers it is known that CBF filters can create undesired stable equilibria on the boundary of the safe set [5, 6], and multi-robot CBF controllers can deadlock [7–9]. When the nominal controller is a

reinforcement-learning (RL) policy, the situation is less clear. The policy may have been trained with the filter in the loop [10–12], it reacts to observations rather than to the true state, and its behaviour near obstacles is learned rather than designed. Recent work shows that training with a sufficiently permissive filter loses no asymptotic performance [13], and that adding a penalty on the filter's correction improves the performance of filter-trained agents in trajectory tracking [10]. To our knowledge, no published study characterises how learned navigation policies deadlock behind a filter, or how the training setup changes the deadlock mechanism. This statement rests on a targeted literature search; a systematic search is listed as a remaining gap.

This paper studies that question in a deliberately simple and fully reproducible setting. The robot is a unicycle, the obstacles are static circles, and the filter is a one-step discrete-time CBF on the speed [14]. The setting is simple enough that the filter's behaviour can be analysed exactly, and that every timeout can be checked against a geometric feasibility oracle.

Research questions.

RQ1 Which properties of a one-step discrete-time CBF speed filter determine where and how a filtered robot can stall?

RQ2 By which mechanisms do learned policies deadlock behind such a filter, and does training with the filter change the mechanism?

RQ3 Which simple remedies (stochastic deployment, the published correction penalty, heading-level modification) remove which mechanism? **[EVIDENCE REQUIRED:** Stochastic deployment and a test-time heading modification have been evaluated; the correction penalty (E1) has not.]

Contributions.

1. A geometric characterisation of the one-step speed filter for the unicycle (Section 4). One-step safety is independent of the commanded turn rate (Proposition 1), so heading-level remedies can be added without weakening the guarantee. The set of headings that permit motion collapses as the barrier approaches zero (Proposition 2), and a stall bound follows (Corollary 1). Proposition 3 shows that self-stops cannot be removed by any change of the speed limit.

2. An evaluation protocol for deadlocks in filtered navigation. It combines a geometric feasibility oracle, a stall taxonomy (filter-blocked vs. self-stop) and uncertainty estimates over both training seeds and test scenarios (Section 5).

3. Empirical evidence from ten PPO agents and two classical controllers (Section 6). The two stall mechanisms exist, and they depend on the training setup. They respond differently to stochastic deployment and to a heading modification. They replicate on previously unused scenarios. The filter-blocked stalls match the geometry predicted by Proposition 2, and self-stops are not an artefact of early checkpoint selection.

We do not claim a new safety filter or a new training algorithm. The safety proof of the filter is standard [14] and is restated only for completeness.

2 Related work

Safety filters and CBFs. CBF quadratic programs [3, 4] and their discrete-time form [14, 15] are the standard tools for enforcing state constraints around arbitrary nominal controllers. High-order CBFs handle constraints with higher relative degree [16]. Predictive and reachability-based filters offer less conservative alternatives at higher computational cost [1, 17]. Robust, measurement-robust and stochastic variants address model and state uncertainty [18–21].

Undesired equilibria and deadlock. Reis et al. [5] showed that CBF-QPs can introduce asymptotically stable equilibria on the safe-set boundary, and Tan and Dimarogonas [6] characterised them in general. In multi-robot navigation, deadlock was analysed as a force equilibrium on the boundary [7, 22] and resolved through instability of the deadlock set [8] or with discrete-time CBFs [9]. All of these analyses assume a model-based nominal controller. Potential fields have the analogous problem of local minima [23]. Reach-avoid RL gives safety and liveness guarantees by learning a reach-avoid value function [24], a different design from post-hoc filtering.

Safety filters and shields in RL. Shields replace unsafe actions during learning [25, 26], and CBF filters have been combined with RL in several ways [27–30]. Krasowski et al. [12] compare action replacement, projection and masking, and report that a penalty on interventions helps. Thumm et al. [31] modify the agent's action proactively to reduce fail-safe interventions. Pizarro Bejarano et al. [10] study filtering during training, a correction penalty and safe resets for PPO on quadrotor trajectory tracking. Yang et al. [11] filter rollouts and add a CBF reward term so that the policy can be deployed without a filter. Oh et al. [13] prove that a sufficiently permissive filter does not degrade asymptotic optimality. CBF-based safe RL for navigation includes [32–35]. Barriers synthesised online from range data remove the known-map assumption [36, 37].

Evaluation practice. RL results vary strongly between seeds [38], and few-run studies need interval estimates rather than point estimates [39]. Navigation benchmarks such as BARN [40] and Arena-Rosnav [41], and community guidelines [42], stress classical baselines and standardised metrics.

Table 1 positions this work. Entries for other papers reflect the parts we read (see LITERATURE_REVIEW.md for the reading depth); "n.r." means not reported in those parts, not that the paper lacks the property.

3 Problem formulation

Robot and environment. The robot state is $x_t = (p_t, \theta_t)$ with position $p_t \in [0, W]^2$, $W = 10\,\mathrm{m}$, and heading θ_t. The command is $u_t = (v_t, \omega_t) \in \mathcal{U} = [0, v_{\max}] \times [-\omega_{\max}, \omega_{\max}]$ with $v_{\max} = 1\,\mathrm{m/s}$ and $\omega_{\max} = 1\,\mathrm{rad/s}$. The simulator integrates

$$p_{t+1} = p_t + v_t\,\Delta t\, e(\theta_t), \qquad \theta_{t+1} = \theta_t + \omega_t \Delta t, \qquad e(\theta) = (\cos\theta, \sin\theta)^\top, \tag{1}$$

with $\Delta t = 0.1\,\mathrm{s}$. Obstacles are static discs with centres c_i and radii $R_i \in [0.3, 0.8]\,\mathrm{m}$; the robot is a disc of radius $r = 0.3\,\mathrm{m}$.

Barriers and filter. For a safety margin d (0.3 m) and a wall margin m (0.1 m), define

$$\begin{aligned} h_i(p) &= \|p - c_i\| - (R_i + r + d) && \text{(obstacle } i\text{)}, \\ h_W(p) &\in \{p_x - (r+m),\ W - p_x - (r+m),\ p_y - (r+m),\ W - p_y - (r+m)\} && \text{(walls)}. \end{aligned} \tag{2}$$

Table 1: Positioning. Y = yes; – = no; n.r. = not reported in the parts read; (Y) = partly.

Work	Learned policy	Filter in training	Navigation	Deadlock / liveness	Equilibria of learned policy	Feasibility oracle
Reis et al. [5]; Tan & Dimarogonas [6]	–	–	(Y)	Y	–	–
Grover et al. [7]; Jankovic et al. [8]; Chandra et al. [9]	–	–	Y	Y	–	n.r.
Krasowski et al. [12]	Y	Y	–	n.r.	–	–
Thumm et al. [31]	Y	Y	(Y)	n.r.	–	n.r.
Pizarro Bejarano et al. [10]	Y	Y	–	chattering	–	–
Oh et al. [13]	Y	Y	n.r.	n.r.	–	–
Yang et al. [11]	Y	Y	Y	n.r.	n.r.	n.r.
Marzari et al. [33]; Guo et al. [35]	Y	(Y)	Y	–	–	–
This work	Y	Y	Y	Y	Y	Y

The safe set is $\mathcal{S} = \{p : h_k(p) \geq 0 \ \forall k\}$. The closing rate per unit speed is $a_i(p,\theta) = e(\theta)^\top (c_i - p)/\|c_i - p\|$ for an obstacle, and $-\cos\theta, \cos\theta, -\sin\theta, \sin\theta$ for the four walls. With $\gamma = 0.5\,\mathrm{s}^{-1}$ (so $\gamma\Delta t = 0.05 \leq 1$), the filter returns

$$F(x,u) = \big(\min\{\mathrm{clip}(v), \bar{v}(x)\},\ \mathrm{clip}(\omega)\big), \qquad \bar{v}(x) = \min\Big\{v_{\max},\ \min_{k:\,a_k>0} \frac{\gamma\, h_k(p)}{a_k(p,\theta)}\Big\}. \tag{3}$$

Because every constraint bounds the single variable v, (3) is the exact solution of $\min_v (v - v^{\mathrm{nom}})^2$ subject to $a_k v \leq \gamma h_k$ and $0 \leq v \leq v_{\max}$.

Policies. A policy maps an observation o_t to a nominal command. The observation contains the noisy normalised position, $\sin\theta$ and $\cos\theta$, goal distance and bearing, a proximity feature, and 16 range readings up to 3 m. The closed loop is $x_{t+1} = f(x_t, F(x_t, \pi(o_t)))$. An episode succeeds when the goal is within 0.5 m; it ends with a collision if the robot body touches an obstacle or leaves the workspace; otherwise it times out after 200/300/400 steps (beginner/intermediate/expert).

Definition 1 (Stall classes)**.** A timeout is a *stall* if the net displacement over its last $K = 100$ steps is below $\delta = 0.10$ m. A stall is (i) *filter-blocked* (Type I) if the filter reduced the commanded speed in at least half of those steps; (ii) a *self-stop* (Type II) if, in at least half of them, the policy itself commanded a speed of at most 10^{-3} m/s; (iii) *other* otherwise. A timeout that is not a stall is *moving*.

4 Analysis of the one-step speed filter

Lemma 1 (Forward invariance [14])**.** *If $p_0 \in \mathcal{S}$ and $u_t = F(x_t, \cdot)$ for all t, then $h_k(p_{t+1}) \geq (1 - \gamma\Delta t) h_k(p_t)$ for every barrier k. In particular, $h_k(p_t) > 0$ for all t if $h_k(p_0) > 0$.*

Proof. By convexity of the norm, $\|p_{t+1} - c_i\| \geq \|p_t - c_i\| - \Delta t\, v_t\, a_i(p_t, \theta_t)$. If $a_i > 0$, the filter enforces $v_t a_i \leq \gamma h_i(p_t)$; if $a_i \leq 0$, the barrier cannot decrease. Wall barriers are affine in p, so the same bound holds with equality. Strict positivity follows from $\gamma\Delta t < 1$. □

Proposition 1 (Heading freedom)**.** *Let $\psi : \mathcal{X} \times \mathcal{U} \to [-\omega_{\max}, \omega_{\max}]$ be any map, and let $F_\psi(x,u) = (\min\{\mathrm{clip}(v), \bar{v}(x)\}, \psi(x,u))$. Then Lemma 1 holds for F_ψ.*

Proof. By (1), p_{t+1} depends only on (p_t, θ_t, v_t), and $\bar{v}(x_t)$ depends only on (p_t, θ_t). The turn rate ω_t affects only θ_{t+1}, which the filter evaluates again at step $t+1$. The proof of Lemma 1 therefore applies at every step, whatever ψ returns. □

Remark 1. Proposition 1 has two consequences. First, the one-step filter can never *steer* the robot away from an obstacle: only the policy, or a heading modification ψ, can do so. Second, heading-level remedies for deadlock, such as a tangential turn when the speed limit is active, inherit the safety guarantee without further proof. In continuous time this corresponds to the position barrier having relative degree two with respect to ω [16].

Proposition 2 (Collapse of the admissible headings)**.** *Fix p with $0 \le h_i(p) \le \varepsilon$ for an obstacle i, and a speed $\underline{v} > 0$. Every heading θ with $\bar{v}(p, \theta) \ge \underline{v}$ satisfies $a_i(p, \theta) \le \gamma\varepsilon/\underline{v}$; that is, the angle between $e(\theta)$ and the direction to c_i is at least* $\arccos(\min\{1, \gamma\varepsilon/\underline{v}\})$*. As $\varepsilon \to 0$, this set of headings approaches the closed half-circle $\{\theta : a_i(p, \theta) \le 0\}$.*

Proof. If $a_i(p, \theta) > 0$, then $\bar{v} \le \gamma h_i / a_i \le \gamma\varepsilon/a_i$, and $\bar{v} \ge \underline{v}$ requires $a_i \le \gamma\varepsilon/\underline{v}$. If $a_i \le 0$, the inequality holds trivially. □

Corollary 1 (Stall bound)**.** *If, over K consecutive steps, $h_i(p_t) \le \varepsilon$ and $a_i(p_t, \theta_t) \ge \bar{a} > 0$, then the robot moves at most $K\Delta t\,\gamma\varepsilon/\bar{a}$, whatever speed the policy commands. For a head-on approach ($a_i \equiv 1$) at the speed limit, $h_i(p_{t+1}) = (1 - \gamma\Delta t)\, h_i(p_t)$ holds with equality, so the barrier decays geometrically and the robot approaches, but never reaches, the boundary.*

Proof. The displacement per step is $v_t \Delta t \le \bar{v}\Delta t \le \gamma\varepsilon\Delta t/\bar{a}$. For straight motion toward the centre, the convexity bound in Lemma 1 is tight, giving the stated equality. □

Proposition 3 (Self-stops do not depend on the speed limit)**.** *Let F' be any filter whose executed speed satisfies $0 \le v^{\text{exec}} \le \text{clip}(v^{\text{nom}})$. At a state where the deterministic policy commands $\pi_v(o) \le 0$, the executed speed is 0 under F' and without any filter. Hence no change of the speed limit can remove a Type II stall; only a change of the heading (Proposition 1) or of the policy can.*

Proof. $\text{clip}(\pi_v(o)) = 0$, so $0 \le v^{\text{exec}} \le 0$. □

Propositions 2 and 3 give testable predictions. Filter-blocked stalls should occur at $h \approx 0$ with headings within 90° of the obstacle direction. Self-stops should occur at $h > 0$, and they should be unaffected by the filter's speed limit.

5 Experimental setup

Agents. PPO [43] with separate 256×256 tanh actor and critic networks, a diagonal Gaussian policy whose samples are clipped by the environment, generalised advantage estimation and bootstrapping at time-limit truncations. Training ran for 10^6 steps over a three-stage curriculum (beginner, intermediate, expert; advance at 70 % rolling success). Five seeds (0–4) were trained without the filter and five with the filter in the loop. In the filtered runs, the buffer stores the policy's own sample, as in [10]. The checkpoint with the highest success on 60 validation episodes (20 per stage), on seeds disjoint from the test seeds, was kept. One hyperparameter set was used for both variants and was not tuned for the filtered setting. All ten runs reached the expert stage within 43k–76k steps.

Baselines. A go-to-goal heading controller, which ignores obstacles, and a hand-tuned artificial potential field on the range readings [23]. **[EVIDENCE REQUIRED:** Published baselines are missing: correction-penalty PPO [10] (E1), PPO-Lagrangian [44], and a classical local planner such as DWA [45] or MPC with a DT-CBF [15].]

Test scenarios. 100 seeded scenarios per stage (seed bases $2 \cdot 10^6 + 10^5 s$). The expert stage has 8–12 obstacles. A grid oracle (5 cm, 8-connected Dijkstra on the free space inflated by $r + d$) found a margin-keeping path in all 300 scenarios. Two properties of the generator must be kept in mind:

- The goal is clipped to $[1.5, 8.5]^2$, so 53 of the 100 expert goals are closer than the configured minimum of 5 m (maximum 8.9 m).
- Three scenarios (two beginner, one expert) start inside the goal region and count as successes at step 1 for every controller.

The 0.3 m margin and the sampled-action evaluation were chosen after inspecting these test scenarios, so the test set is not a clean held-out set for those two choices.

Fresh scenarios and pre-registration. To address this, we evaluated the agents on 500 expert scenarios that had never been used (seeds $9 \cdot 10^6$ to $9 \cdot 10^6 + 499$). The hypotheses, decision rules and analysis were committed to the repository before this evaluation was run. The scenarios come from the same generator, so the goal-clipping caveat still applies. **[EVIDENCE REQUIRED:** Re-evaluation with a corrected generator (E3).]

Heading escape. As a test-time intervention, we add to the filter a turn-rate modification ψ. When the filter reduces the commanded speed, the turn rate becomes $\mathrm{clip}(\omega^{\mathrm{nom}} + \kappa\, s\, \omega_{\max})$. Here $s \in \{-1, +1\}$ is the turn direction that increases the angle between the heading and the direction of the binding barrier (towards its tangent); if that angle is zero, s points to the goal side. By Proposition 1, the safety guarantee holds for every κ. The value $\kappa = 0.5$ was selected from $\{0.5, 1.0\}$ by success on 100 validation scenarios (seeds $1.2 \cdot 10^6$ onward) using a rule fixed in advance; it was then frozen. The agents are not retrained.

Checkpoint selection. To test whether self-stops are an artefact of selecting early checkpoints, we also evaluated the final (10^6-step) checkpoints of the five filter-trained runs on the fresh scenarios. These checkpoints are local training outputs and are not yet in the public repository; their SHA-256 hashes are recorded.

Conditions and classification. Each PPO agent was evaluated with the filter off and on, using the policy mean. Each was additionally evaluated with the filter on and actions sampled from the policy, seeded per episode. Every timeout with the filter on was replayed deterministically from its seed and classified by Definition 1. This classification is a post-hoc analysis made after the benchmark had been run.

Statistics. For PPO, the unit of replication is a training seed. Seed-level 95 % t-intervals over five seeds describe variation between training runs on the fixed scenario set. Comparisons between conditions use a paired hierarchical bootstrap (10,000 replicates). Scenarios are resampled jointly for both conditions, because all conditions share the same scenarios, and seeds are resampled within each condition. Proportions for the deterministic baselines use Wilson intervals over episodes. A

Table 2: Expert stage, 100 test scenarios. PPO: mean over five training seeds with 95 % t-interval over seeds (500 episodes); baselines: Wilson interval over 100 episodes. Sources: `results/benchmark/summary.csv`, `results/stochastic_eval/summary.json`.

Condition	Success (%)	Collision (%)	Timeout (%)	Interventions (%)
PPO trained w/o filter, filter off	94.6 [91.7, 97.5]	1.0	4.4	–
PPO trained w/o filter, filter on	65.0 [61.5, 68.5]	0	35.0	58.0
PPO trained w/o filter, filter on, sampled	70.0 [66.3, 73.7]	0	30.0	n.r.
PPO trained with filter, filter on	61.0 [55.4, 66.6]	0	39.0	41.5
PPO trained with filter, filter on, sampled	76.4 [68.2, 84.6]	0	23.6	n.r.
Go-to-goal, filter off	57.0 [47.2, 66.3]	43.0	0	–
Go-to-goal, filter on	46.0 [36.6, 55.7]	0	54.0	67.2
Potential field, filter off	73.0 [63.6, 80.7]	0	27.0	–
Potential field, filter on	72.0 [62.5, 79.9]	0	28.0	40.7

two-way decomposition of episode success into seed, scenario and residual sums of squares quantifies the two sources of uncertainty. No multiplicity correction was applied, because the contrasts are reported as estimates with intervals and are not used as a family of hypothesis tests. With five seeds, the bootstrap underestimates seed variance, so the intervals should be read as approximate.

Reproducibility. The repository contains the committed checkpoints, and regenerating the benchmark from them reproduces the per-episode files byte for byte. All analysis scripts read only stored outputs; Section 9 lists them.

6 Results

6.1 Safety and success

With the filter on, none of the 3,600 test episodes (all controllers and stages) had a collision, and the smallest obstacle clearance was the 0.30 m margin (Table 2). The filter's cost in success depends on the controller. For PPO trained without the filter, expert success fell from 94.6 % to 65.0 % when the filter was added at test time, a paired difference of 29.6 points [21.6, 38.4]. The agent trained with the filter reached 61.0 %. The paired difference to the unfiltered-trained agent with the filter on was −4.0 points [−10.8, 3.2] with mean actions, and +6.4 points [−1.6, 15.2] with sampled actions. Neither comparison shows that training with the filter improves success.

6.2 Two stall mechanisms

Figure 1 and Table 3 classify every timeout with the filter on.

- **PPO trained without the filter.** Filter-blocked stalls dominate: 141 of 175 expert timeouts and 68 of 81 intermediate timeouts, with mean actions. In these stalls the policy commands full speed (median 1.0 m/s) while the turn rate changes sign a median of 70 times in 100 steps.
- **PPO trained with the filter.** Self-stops dominate: 122 of 195 expert timeouts and 68 of 100 intermediate timeouts. The policy commands zero speed at a median barrier value of 0.41 m (interquartile range 0.16–0.72 m), well before the keep-out boundary. In 89 of the 122 expert self-stops, the goal lies within 45° of the direction of the nearest obstacle (median 26°), and the robot barely turns (median $|\omega|$ 0.04 rad/s).

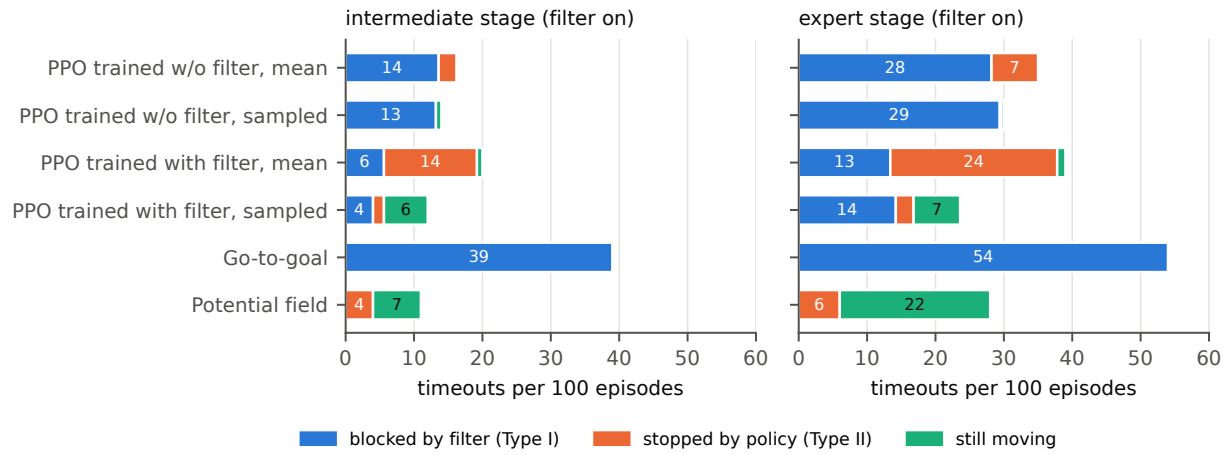


Figure 1: Timeouts per 100 episodes with the filter on, by stall class (Definition 1). PPO: five seeds × 100 scenarios; baselines: 100 scenarios. Source: `research/audit_outputs/stall_taxonomy_all.csv`.

Table 3: Timeout classes with the filter on (counts; PPO over 500 episodes per row, baselines over 100). Source: `research/audit_outputs/stall_taxonomy_all.csv`.

	Intermediate			Expert		
Condition	Type I	Type II	moving	Type I	Type II	moving
PPO trained w/o filter, mean	68	13	0	141	34	0
PPO trained w/o filter, sampled	66	0	4	147	1	2
PPO trained with filter, mean	28	68	4	67	122	6
PPO trained with filter, sampled	20	8	32	71	13	34
Go-to-goal	39	0	0	54	0	0
Potential field	0	4	7	0	6	22

- **Baselines.** The two classical controllers bracket the PPO behaviour. Go-to-goal, which always commands forward speed toward the goal, fails only by filter-blocking (54 of 54 expert timeouts). The potential field never stalls against the filter: it oscillates in local minima (22 "moving" timeouts) or stops (6).

The taxonomy is insensitive to its thresholds. Varying the displacement threshold (0.05, 0.10 and 0.20 m) and the share threshold (30, 50 and 70 %) changes the expert counts by at most 3 (filter-blocked) and 3 (self-stop) for either agent type. Varying the window length K over 50, 100, 150 and 200 steps, with the displacement threshold scaled to the window, leaves the counts of the agent trained without the filter unchanged (141 and 34). For the agent trained with the filter, the filter-blocked count ranges over 63–69 and the self-stop count over 119–122.

6.3 The geometry of filter-blocked stalls matches the analysis

Proposition 2 predicts that a robot blocked at the boundary heads less than 90° from the obstacle direction. The data confirm this (Figure 2a).

- 617 of the 618 filter-blocked stalls of all PPO conditions have a heading below 90°; the remaining one is at 91.3°. The median is 66.8°, so the robots are typically not facing the

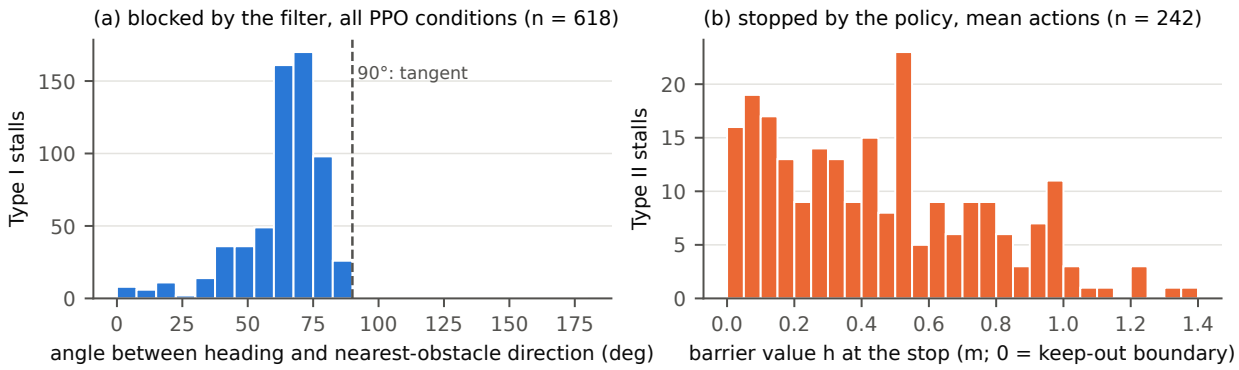


Figure 2: (a) Heading relative to the nearest obstacle at the end of filter-blocked stalls (all PPO conditions, both stages); the dashed line marks 90° (tangent). (b) Barrier value at self-stops (PPO, mean actions). Source: `research/audit_outputs/stall_taxonomy_all.csv`.

obstacle head-on.

- The obstacle barrier at the end of these stalls is at most 4.2 mm, consistent with the geometric decay in Corollary 1.
- In all 618 stalls, the active barrier is an obstacle: no wall barrier is below 1 mm.

Self-stops, in contrast, are spread over barrier values from 0 to 1.4 m (Figure 2b). This is consistent with Proposition 3: these stops are a property of the policy, not of the filter.

6.4 Training with the filter shifts the mechanism; sampling removes self-stops

Paired comparisons (Figure 3) quantify both effects.

- **Training with the filter.** On the expert stage, it lowered the filter-blocked rate by 14.8 points [5.6, 24.4] and raised the self-stop rate by 17.6 points [8.2, 27.0]. On the intermediate stage, the corresponding changes were −8.0 [−14.8, −1.8] and +11.0 [5.6, 17.2].
- **Sampled actions** (agent trained with the filter). The expert self-stop rate fell by 21.8 points [12.4, 31.6], while the filter-blocked rate did not change (+0.8 [−7.6, 10.0]).
- **Success.** Sampling raised expert success by 15.4 points [7.8, 24.0] for the filter-trained agent, against 5.0 [0.4, 10.4] for the agent trained without the filter.

Stochastic deployment therefore acts on the learned mechanism only. The filter-blocked stalls persist, as Proposition 2 suggests they should for any policy that keeps a heading within 90° of the obstacle.

6.5 Sources of uncertainty

The decomposition of expert success shows how much the test scenario matters. Depending on the condition, the scenario term accounts for 25–86 % of the sum of squares, against under 2 % for the seed term. Seed-only intervals such as those in Table 2 therefore describe reproducibility of training on these 100 scenarios; they understate the uncertainty about the scenario distribution. An approximate fit of interval width to the number of seeds and scenarios suggests that about 10 seeds and 1,000 scenarios per stage would give half-widths near 4–5 points for differences of this kind. Beyond about 1,000 scenarios, more seeds help more than more scenarios.

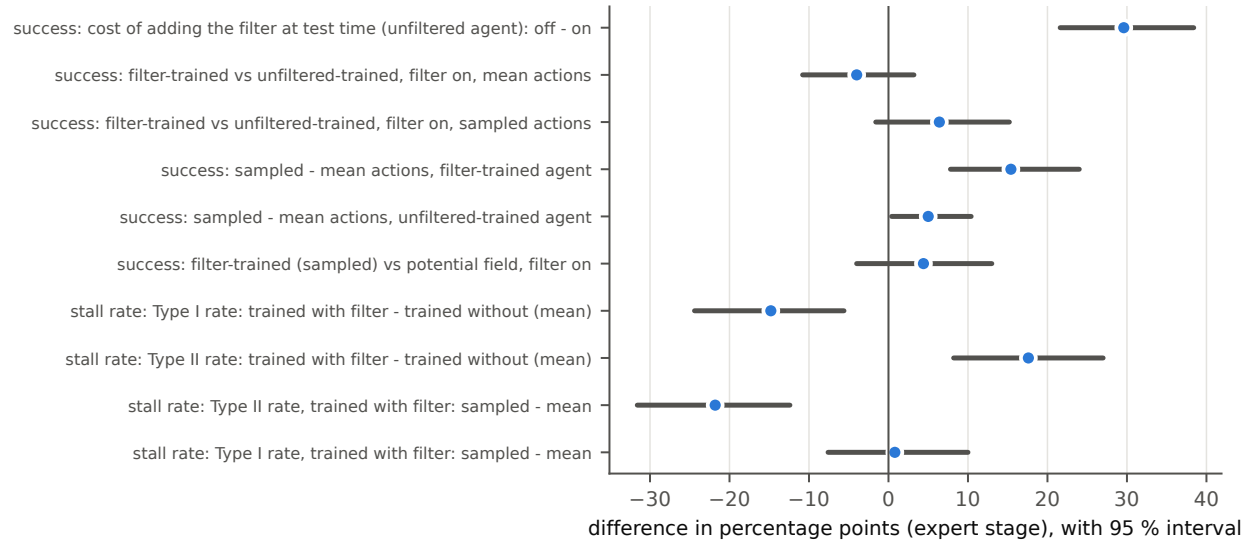


Figure 3: Paired differences (expert stage) with 95 % paired hierarchical bootstrap intervals. Success differences: `research/audit_outputs/paired_comparisons_expert.csv`; stall-rate differences: `paper_draft/generated/stats.json`.

Table 4: Pre-registered evaluation on 500 previously unused expert scenarios (five agents per row, 2,500 episodes, mean actions). κ: heading escape (0 = plain filter). Source: `research/experiments/lightweight/results/summary.json`.

Agents	κ	Success (%)	Timeouts	Type I	Type II	moving
trained w/o filter	0	68.3	792	709	81	2
trained w/o filter	0.5	75.6	611	523	85	3
trained with filter	0	63.9	903	297	577	29
trained with filter	0.5	64.0	901	244	615	42
trained with filter, final checkpoints	0	59.1	1023	123	870	30

6.6 Replication on previously unused scenarios

On the 500 fresh expert scenarios (5,000 episodes per condition, mean actions), the taxonomy and both training contrasts replicate (Table 4).

- Filter-blocked stalls make up 709 of the 792 timeouts of the agents trained without the filter.
- Self-stops make up 577 of the 903 timeouts of the agents trained with it.
- Training with the filter lowered the filter-blocked rate by 16.5 points [10.1, 22.8] and raised the self-stop rate by 19.8 points [12.8, 26.1].

Both effects are close to those on the original test set. In this larger sample, expert success of the filter-trained agents was also lower than that of the agents trained without the filter, by 4.4 points [0.2, 8.5]. This comparison was exploratory, not one of the pre-registered tests. There were no collisions in the 10,000 fresh-scenario episodes, and the minimum clearance was 0.30 m.

6.7 Self-stops are not an artefact of checkpoint selection

The final checkpoints of the filter-trained runs have more self-stops than the selected checkpoints: +11.7 points [3.0, 20.8]. They have fewer filter-blocked stalls (−7.0 [−13.0, −1.5]) and lower success (−4.8 [−8.8, −1.2]). The increase comes from the three seeds whose selected checkpoint was from early in training (0.1–0.3M steps). Self-stops in these seeds rose from 139, 51 and 105 to 205, 174 and 206. In the seed whose best checkpoint is its final one, the results are identical. Longer training with the filter therefore strengthens the self-stopping behaviour rather than removing it.

6.8 Remedies

Heading escape (test time, frozen agents). With $\kappa = 0.5$ (Table 4), the filter-blocked rate of the agents trained without the filter fell by 7.4 points [4.8, 10.2], and their success rose by 7.2 points [5.0, 9.7]. The reduction held in every seed. Self-stops of the filter-trained agents did not change (+1.5 [−0.8, 3.9]), nor did their success (+0.1 [−2.2, 2.4]). This is expected. The escape is triggered only when the filter reduces the commanded speed, which never happens while the policy commands zero speed (Proposition 3). A heading change triggered during a stop could in principle change the policy's observation and release it, but we did not test this. There were no collisions, as Proposition 1 guarantees. The effect is real but small. It is below the 10-point threshold we fixed in advance, and 523 of the remaining 611 timeouts of these agents are still filter-blocked. A one-step heading rule is therefore not enough to resolve Type I stalls; it is a lower bound on what filter-level heading changes can do.

Correction penalty. [**EVIDENCE REQUIRED:** E1: correction-penalty PPO [10] with $\alpha = 1$ against an identical control, evaluated on the fresh scenarios. This needs training; a one-seed pilot (about 20 min) is proposed in minimum_experiments.md.]

7 Discussion

Two different problems. Most discussions of filter-induced deadlock treat it as one problem of the filter [5, 6]. Our data show two different problems, and which one dominates depends on the training setup. A policy that never saw the filter keeps pushing at the boundary. It is held there because, by Proposition 2, no heading within 90° of the obstacle admits motion. This is the learned-policy counterpart of the undesired boundary equilibria of CBF-QPs. A policy trained with the filter mostly avoids being blocked, but it has learned to stop on its own when an obstacle lies between it and the goal. By Proposition 3, no change of the speed limit can affect such stops.

Why do filter-trained policies stop? We have no causal evidence yet. A plausible mechanism is *action aliasing* in the filtered environment. Every sampled speed $v \leq 0$ is executed as 0, and every speed above the limit is executed as the limit. Within these sets, the executed action, and therefore the return, does not depend on the sampled value, so the policy-gradient signal cannot distinguish these actions. The mean can then drift into the stopping region near obstacles, where forward commands were often truncated during training. Sampled actions occasionally leave this region, which is consistent with the observed reduction of self-stops. The final checkpoints have more self-stops than the early selected ones, which fits a behaviour that the training process reinforces. It does not tell us whether aliasing or the reward is responsible. The non-potential-based risk penalty in the reward is an alternative or additional explanation.

Hypothesis 1. Removing the aliasing (a bounded action distribution, or a policy parameterised in the executed-action space) removes Type II stops, while a reward-only change does not. **[EVIDENCE REQUIRED:** Untested; minimum_experiments.md, E4.]

Relation to theory on filtered training. Oh et al. [13] show that a sufficiently permissive filter does not limit the asymptotically optimal return. The speed filter here permits turning in place, and the oracle shows that a margin-keeping path exists in every test scenario. A large part of the remaining gap is therefore plausibly a learning problem rather than a limit imposed by the filter. Checking the formal permissiveness condition for this filter is left open.

Implications for remedy design. Proposition 1 means that heading-level remedies come at no cost to the one-step guarantee. The simple escape tested here shows that such a change helps, but only modestly. A heading rule that looks one step ahead cannot plan a way around an obstacle, so multi-step or planning-level heading changes are the natural next candidates. Proposition 3 means that any remedy acting only on the speed limit (for example a less conservative γ) cannot remove self-stops. Heading changes that are triggered only by speed reductions cannot remove them either, as the evaluation confirmed. The published correction penalty [10] penalises both the filter's truncation and the box clipping of $v \leq 0$. It may therefore act on both mechanisms, which makes it the most important missing baseline.

8 Limitations and threats to validity

- **External validity.** 2D simulation with exact Euler kinematics, static circular obstacles, exact state for the filter and a known map. Nothing was tested on a robot, and no finite simulation establishes real-world safety. The filter's guarantee holds at the sampled states. Between samples, the chord bound limits the penetration to about 1.4 mm for the generated obstacle sizes; continuous-time motion is not covered.
- **Scenario design.** Goal clipping shortens many goal distances. Three trivial scenarios add 1–2 points to every success rate in their stages. The layouts are fairly open: the median oracle path is 6–8 % longer than the straight line.
- **Test-set reuse.** The margin, the sampled-action evaluation and the stall analysis were all developed on the same test scenarios. The pre-registered evaluation on 500 unused scenarios reduces, but does not remove, this threat. Those scenarios come from the same generator, and only the expert stage, mean actions and one value of κ were tested. The sampled-action variant of that evaluation and the final checkpoints of the agents trained without the filter were planned but not run.
- **Statistical power.** Five seeds per variant, with 100 scenarios per stage on the original test set and 500 on the fresh set. The intervals are approximate, and there is no multiplicity correction beyond the pre-registered decision rules.
- **Artefacts.** The final checkpoints used for the checkpoint-selection analysis are not yet in the public repository.
- **Construct validity of the taxonomy.** The class thresholds are heuristic, but varying the displacement threshold, the share threshold and the window length changed the expert counts by at most 4.

- **Baselines and tuning.** PPO hyperparameters were not tuned for the filtered setting, the potential field was tuned by hand, and published safe-RL and classical baselines are missing.
- **Privileged feature.** The proximity feature in the observation is computed from the true map.

9 Conclusion

A one-step discrete-time CBF filter makes learned navigation policies collision-free in simulation, but it does not make them live. We showed that the filter's one-step guarantee is independent of the turn rate, and that the headings admitting motion collapse to a half-plane at the boundary. We also showed that learned policies deadlock behind the filter by two distinct mechanisms. A policy trained without the filter is blocked by the filter at the boundary, while a policy trained with the filter mostly stops itself before it. Stochastic deployment removes the second mechanism but not the first. A test-time heading escape reduces the first modestly and leaves the second untouched. Longer training with the filter makes the second more frequent. These results separate what a filter design can fix from what training must fix. [**EVIDENCE REQUIRED:** Whether the published correction penalty resolves either mechanism (E1) is the main open question.]

Data and code availability

Code, committed checkpoints, per-episode results and analysis scripts: `https://github.com/pouyapd/SafeNav-RL`. Benchmark: `results/benchmark/`; sampled-action evaluation: `results/stochastic_eval/`; stall taxonomy: `research/scripts/stall_taxonomy.py`; figures and statistics in this paper: `research/scripts/paper_figures_stats.py`; pre-registered fresh-scenario evaluation, heading escape and checkpoint comparison: `research/experiments/lightweight/` (protocol, runner, analysis and per-episode results). [**EVIDENCE REQUIRED:** Archive a tagged release with a DOI (e.g. Zenodo) before submission.]

References

[1] Kai-Chieh Hsu, Haimin Hu, and Jaime F. Fisac. The safety filter: A unified view of safety-critical control in autonomous systems. *Annual Review of Control, Robotics, and Autonomous Systems*, 7(1):47–72, 2024. doi: 10.1146/annurev-control-071723-102940.

[2] Lukas Brunke, Melissa Greeff, Adam W. Hall, Zhaocong Yuan, Siqi Zhou, Jacopo Panerati, and Angela P. Schoellig. Safe learning in robotics: From learning-based control to safe reinforcement learning. *Annual Review of Control, Robotics, and Autonomous Systems*, 5(1):411–444, 2022. doi: 10.1146/annurev-control-042920-020211.

[3] Aaron D. Ames, Xiangru Xu, Jessy W. Grizzle, and Paulo Tabuada. Control barrier function based quadratic programs for safety critical systems. *IEEE Transactions on Automatic Control*, 62(8):3861–3876, 2017. doi: 10.1109/TAC.2016.2638961.

[4] Aaron D. Ames, Samuel Coogan, Magnus Egerstedt, Gennaro Notomista, Koushil Sreenath, and Paulo Tabuada. Control barrier functions: Theory and applications. In *2019 18th European Control Conference (ECC)*, pages 3420–3431, 2019. doi: 10.23919/ECC.2019.8796030.

[5] Matheus F. Reis, A. Pedro Aguiar, and Paulo Tabuada. Control barrier function-based quadratic programs introduce undesirable asymptotically stable equilibria. *IEEE Control Systems Letters*, 5(2):731–736, 2021. doi: 10.1109/LCSYS.2020.3004797.

[6] Xiao Tan and Dimos V. Dimarogonas. On the undesired equilibria induced by control barrier function based quadratic programs. *Automatica*, 159:111359, 2024. doi: 10.1016/j.automatica.2023.111359.

[7] Jaskaran Grover, Changliu Liu, and Katia Sycara. The before, during, and after of multi-robot deadlock. *The International Journal of Robotics Research*, 42(6):317–336, 2023. doi: 10.1177/02783649221074718.

[8] Mrdjan Jankovic, Mario Santillo, and Yan Wang. Multiagent systems with cbf-based controllers: Collision avoidance and liveness from instability. *IEEE Transactions on Control Systems Technology*, 32(2):705–712, 2024. doi: 10.1109/TCST.2023.3324531.

[9] Rohan Chandra, Vrushabh Zinage, Efstathios Bakolas, Peter Stone, and Joydeep Biswas. Deadlock-free, safe, and decentralized multi-robot navigation in social mini-games via discrete-time control barrier functions. *Autonomous Robots*, 49(2), 2025. doi: 10.1007/s10514-025-10194-8.

[10] Federico Pizarro Bejarano, Lukas Brunke, and Angela P. Schoellig. Safety filtering while training: Improving the performance and sample efficiency of reinforcement learning agents. *IEEE Robotics and Automation Letters*, 10(1):788–795, 2025. doi: 10.1109/LRA.2024.3512374.

[11] Lizhi Yang, Blake Werner, Massimiliano de Sa, and Aaron D. Ames. Cbf-rl: Safety filtering reinforcement learning in training with control barrier functions. In *2026 IEEE International Conference on Robotics and Automation (ICRA)*, pages 17825–17832, 2026. doi: 10.1109/ICRA57385.2026.11695808.

[12] Hanna Krasowski, Jakob Thumm, Marlon Müller, Lukas Schäfer, Xiao Wang, and Matthias Althoff. Provably safe reinforcement learning: Conceptual analysis, survey, and benchmarking. *Transactions on Machine Learning Research*, 2023. No DOI; arXiv:2205.06750.

[13] Donggeon David Oh, Duy P. Nguyen, Haimin Hu, and Jaime Fernández Fisac. Provably optimal reinforcement learning under safety filtering. *Proceedings of IASEAI Conference*, 2(1):488–500, 2026. doi: 10.1609/iaseai.v2i1.43046.

[14] Ayush Agrawal and Koushil Sreenath. Discrete control barrier functions for safety-critical control of discrete systems with application to bipedal robot navigation. In *Robotics: Science and Systems XIII*, 2017. doi: 10.15607/RSS.2017.XIII.073.

[15] Jun Zeng, Bike Zhang, and Koushil Sreenath. Safety-critical model predictive control with discrete-time control barrier function. In *2021 American Control Conference (ACC)*, pages 3882–3889, 2021. doi: 10.23919/ACC50511.2021.9483029.

[16] Wei Xiao and Calin Belta. High-order control barrier functions. *IEEE Transactions on Automatic Control*, 67(7):3655–3662, 2022. doi: 10.1109/TAC.2021.3105491.

[17] Kim Peter Wabersich and Melanie N. Zeilinger. A predictive safety filter for learning-based control of constrained nonlinear dynamical systems. *Automatica*, 129:109597, 2021. doi: 10.1016/j.automatica.2021.109597.

[18] Shishir Kolathaya and Aaron D. Ames. Input-to-state safety with control barrier functions. *IEEE Control Systems Letters*, 3(1):108–113, 2019. doi: 10.1109/LCSYS.2018.2853698.

[19] Ryan K. Cosner, Andrew W. Singletary, Andrew J. Taylor, Tamas G. Molnar, Katherine L. Bouman, and Aaron D. Ames. Measurement-robust control barrier functions: Certainty in safety with uncertainty in state. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 6286–6291, 2021. doi: 10.1109/IROS51168.2021.9636584.

[20] Ryan Cosner, Preston Culbertson, Andrew Taylor, and Aaron Ames. Robust safety under stochastic uncertainty with discrete-time control barrier functions. In *Robotics: Science and Systems XIX*, 2023. doi: 10.15607/RSS.2023.XIX.084.

[21] Ersin Daş, Rahal Nanayakkara, Xiao Tan, Ryan M. Bena, Joel W. Burdick, Paulo Tabuada, and Aaron D. Ames. Safe navigation under state uncertainty: Online adaptation for robust control barrier functions. *IEEE Robotics and Automation Letters*, 11(3):3398–3405, 2026. doi: 10.1109/LRA.2026.3653366.

[22] Jaskaran Singh Grover, Changliu Liu, and Katia Sycara. Deadlock analysis and resolution for multi-robot systems. In *Springer Proceedings in Advanced Robotics*, pages 294–312. Springer, 2021. doi: 10.1007/978-3-030-66723-8_18.

[23] Y. Koren and J. Borenstein. Potential field methods and their inherent limitations for mobile robot navigation. In *Proceedings. 1991 IEEE International Conference on Robotics and Automation*, pages 1398–1404, 1991. doi: 10.1109/ROBOT.1991.131810.

[24] Kai-Chieh Hsu*, Vicenç Rubies-Royo*, Claire Tomlin, and Jaime Fisac. Safety and liveness guarantees through reach-avoid reinforcement learning. In *Robotics: Science and Systems XVII*, 2021. doi: 10.15607/RSS.2021.XVII.077.

[25] Mohammed Alshiekh, Roderick Bloem, Rüdiger Ehlers, Bettina Könighofer, Scott Niekum, and Ufuk Topcu. Safe reinforcement learning via shielding. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 2018. doi: 10.1609/aaai.v32i1.11797.

[26] Steven Carr, Nils Jansen, Sebastian Junges, and Ufuk Topcu. Safe reinforcement learning via shielding under partial observability. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(12):14748–14756, 2023. doi: 10.1609/aaai.v37i12.26723.

[27] Richard Cheng, Gábor Orosz, Richard M. Murray, and Joel W. Burdick. End-to-end safe reinforcement learning through barrier functions for safety-critical continuous control tasks. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(01):3387–3395, 2019. doi: 10.1609/aaai.v33i01.33013387.

[28] Yousef Emam, Gennaro Notomista, Paul Glotfelter, Zsolt Kira, and Magnus Egerstedt. Safe reinforcement learning using robust control barrier functions. *IEEE Robotics and Automation Letters*, 10(3):2886–2893, 2025. doi: 10.1109/LRA.2022.3216996.

[29] Wei Xiao, Tsun-Hsuan Wang, Ramin Hasani, Makram Chahine, Alexander Amini, Xiao Li, and Daniela Rus. Barriernet: Differentiable control barrier functions for learning of safe robot control. *IEEE Transactions on Robotics*, 39(3):2289–2307, 2023. doi: 10.1109/TRO.2023.3249564.

[30] Tu-Hoa Pham, Giovanni De Magistris, and Ryuki Tachibana. Optlayer - practical constrained optimization for deep reinforcement learning in the real world. In *2018 IEEE International*

Conference on Robotics and Automation (ICRA), pages 6236–6243, 2018. doi: 10.1109/ICRA.2018.8460547.

[31] Jakob Thumm, Guillaume Pelat, and Matthias Althoff. Reducing safety interventions in provably safe reinforcement learning. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 7515–7522, 2023. doi: 10.1109/IROS55552.2023.10342464.

[32] Lixing Song, Luke Ferderer, and Shaoen Wu. Safe reinforcement learning for lidar-based navigation via control barrier function. In *2022 21st IEEE International Conference on Machine Learning and Applications (ICMLA)*, pages 264–269, 2022. doi: 10.1109/ICMLA55696.2022.00045.

[33] Luca Marzari, Francesco Trotti, Enrico Marchesini, and Alessandro Farinelli. Designing control barrier function via probabilistic enumeration for safe reinforcement learning navigation. *IEEE Robotics and Automation Letters*, 10(10):9630–9637, 2025. doi: 10.1109/LRA.2025.3596431.

[34] Junjun Xie, Shuhao Zhao, Liang Hu, and Huijun Gao. Certificated actor-critic: Hierarchical reinforcement learning with control barrier functions for safe navigation. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 5482–5488, 2025. doi: 10.1109/ICRA55743.2025.11127225.

[35] Bingbing Guo, Guina Wang, Yiyang Chen, Yue Gao, and Qian Xie. Risk-aware reinforcement learning with dynamic safety filter for collision risk mitigation in mobile robot navigation. *Sensors*, 25(17):5488, 2025. doi: 10.3390/s25175488.

[36] Shaghayegh Keyumarsi, Made Widhi Surya Atman, and Azwirman Gusrialdi. Lidar-based online control barrier function synthesis for safe navigation in unknown environments. *IEEE Robotics and Automation Letters*, 9(2):1043–1050, 2024. doi: 10.1109/LRA.2023.3339059.

[37] Kehan Long, Cheng Qian, Jorge Cortes, and Nikolay Atanasov. Learning barrier functions with memory for robust safe navigation. *IEEE Robotics and Automation Letters*, 6(3):4931–4938, 2021. doi: 10.1109/LRA.2021.3070250.

[38] Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 2018. doi: 10.1609/aaai.v32i1.11694.

[39] Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron Courville, and Marc G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. No DOI; arXiv:2108.13264.

[40] Daniel Perille, Abigail Truong, Xuesu Xiao, and Peter Stone. Benchmarking metric ground navigation. In *2020 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, pages 116–121, 2020. doi: 10.1109/SSRR50563.2020.9292572.

[41] Linh Kästner, Reyk Carstens, Huajian Zeng, Jacek Kmiecik, Teham Bhuiyan, Niloufar Khorsandhi, Volodymyir Shcherbyna, and Jens Lambrecht. Arena-rosnav 2.0: A development and benchmarking platform for robot navigation in highly dynamic environments. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 11257–11264, 2023. doi: 10.1109/IROS55552.2023.10342152.

[42] Anthony Francis, Claudia Pérez-D'Arpino, Chengshu Li, Fei Xia, Alexandre Alahi, Rachid Alami, Aniket Bera, Abhijat Biswas, Joydeep Biswas, Rohan Chandra, Hao-Tien Lewis Chiang, Michael Everett, Sehoon Ha, Justin Hart, Jonathan P. How, Haresh Karnan, Tsang-Wei Edward Lee, Luis J. Manso, Reuth Mirsky, Sören Pirk, Phani Teja Singamaneni, Peter Stone, Ada V. Taylor, Peter Trautman, Nathan Tsoi, Marynel Vázquez, Xuesu Xiao, Peng Xu, Naoki Yokoyama, Alexander Toshev, and Roberto Martín-Martín. Principles and guidelines for evaluating social robot navigation algorithms. *ACM Transactions on Human-Robot Interaction*, 14(2):1–65, 2025. doi: 10.1145/3700599.

[43] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. Preprint, arXiv:1707.06347.

[44] Adam Stooke, Joshua Achiam, and Pieter Abbeel. Responsive safety in reinforcement learning by PID lagrangian methods. In *International Conference on Machine Learning (ICML)*, volume 119 of *PMLR*, 2020. No DOI.

[45] D. Fox, W. Burgard, and S. Thrun. The dynamic window approach to collision avoidance. *IEEE Robotics & Automation Magazine*, 4(1):23–33, 1997. doi: 10.1109/100.580977.